

Personal Information

Mail Biodiversity, Ecology, and Conservation Group
International Institute for Applied Systems Analysis (IIASA)
Schloßplatz 1
A-2361 Laxenburg, Austria

Phone +43 660 6356603

e-mail talluto@iiasa.ac.at

web <https://ltalluto.github.io>

Languages English (native); French (CEFL B2); German (est. A2)

Education

- 2013 **Ph.D.**, *University of Wyoming*, Laramie, Ecology
Dissertation: *The evolution of landscape structure: Eco-evolutionary dynamics drive spatial variation in serotiny in lodgepole pine*
- 2005 **M.S.**, *University of California*, Irvine, Biological Sciences
Thesis: *Shrub-herb interactions in California's coastal sage scrub: Factors affecting establishment of a native understory and historical invasion by annual grasses*
- 2001 **B.S.**, *California Polytechnic State University*, San Luis Obispo, Ecology and Systematic Biology, *cum laude*

Professional Appointments

- 2025–pres **Senior Research Scholar**, *International Institute for Applied Systems Analysis*, Laxenburg, Austria
- 2020–2025 **University Assistant**, *Department of Ecology, University of Innsbruck*, Innsbruck, Austria
- 2018–2019 **Postdoctoral Researcher**, *IGB, Leibniz Institute for Freshwater Ecology and Inland Fisheries*, Berlin, Germany
- 2015–2018 **Postdoctoral Researcher**, *CNRS, Laboratoire d'Écologie Alpine*, Grenoble, France
- 2013–2015 **Postdoctoral Fellow**, *Département de Biologie, Université du Québec*, Rimouski, Québec, Canada
- 2009–2013 **Graduate Research Assistant**, *Department of Zoology & Physiology, University of Wyoming*, Laramie, WY, USA
- 2008–2009 **Lead Biologist**, *Western Riverside County MSHCP Biological Monitoring Program*, Riverside, CA, USA
- 2006–2008 **Avian Program Lead**, *Western Riverside County MSHCP Biological Monitoring Program*, Riverside, CA, USA

Funding

2020 **DRYvER: Securing biodiversity, functional integrity and ecosystem services in DRYing rivER networks**, Horizon 2020: Grant agreement ID: 869226, doi: 10.3030/869226, **€6.6 million**

Role: co-author, work package 3; project participant; co-author 2 on two deliverables.

Publications

31. Fuß, T., Thuile Bistarelli, L., **Talluto, L.**, del Campo, R., Altermatt, F., Masese, F. & Singer, G. (2026) River size drives community assembly processes in river networks. *Global Ecology and Biogeography*, **35**.
30. Martini, J., Fuß, T., Brasseur, M. V., Oester, R., **Talluto, L.**, Walther, F., Wilfling, O., Zizka, V. M. A., Singer, G. & Vitecek, S. (2026) Seasonal Metacommunity Processes of Benthic Invertebrates in the Vjosa/Aoos River. *Ecology and Evolution*, **16**, e73307.
29. Lucas, P. M., Thuiller, W., **Talluto, L.** & 80 additional authors (2025) Trophic interactions are key to understanding the effects of global change on the distribution and functional role of the brown bear. *Global Change Biology*, **31**, e70252.
28. Sinnott-Armstrong, N., Forsythe, D., Benoit, J. M., Chappell, C. R., Coe, L. S. Y., Oliveira, B. F. R. de, Evans, N., Fagre, A. C., Gilligan, J. M., Hamilton, M., Henneberry, C. M., Ishaq, S. L., Johnston, J., Krichilsky, E., Lopez, J. A., McMonigal, K., Campa, M. O. A. de la, Rahman, R., Schwartz, N. E., **Talluto, L.**, Taylor, E. J., Vargas-Muñiz, J. M. & Weissman, J. L. (2025) Protect transgender scientists. *Science [letter]*, **388**, 1283–1284.
27. Fuß, T., Thuile Bistarelli, L., Walther, F., Vitecek, S., **Talluto, L.** & Singer, G. (2024) Geodiversity of a European river network controls algal biodiversity and function. *Communications Earth & Environment*, **5**, 315.
26. Pereira, H. M. et al. (2024) Global trends and scenarios for terrestrial biodiversity and ecosystem services from 1900 to 2050. *Science*, **384**, 458–465 [**L. Talluto** is author 51 of 57].
25. Rittweg, T. D., Trueman, C., Wiedenbeck, M., Fietzke, J., Wolter, C., **Talluto, L.**, Dennenmoser, S., Nolte, A. & Arlinghaus, R. (2024) Variable habitat use supports fine-scale population differentiation of a freshwater piscivore (northern pike, *Esox lucius*) along salinity gradients in brackish lagoons. *Oecologia*.
24. **Talluto, L.**, del Campo, R., Estévez, E., Altermatt, F., Datry, T. & Singer, G. (2024) Towards (better) fluvial meta-ecosystem ecology: a research perspective. *npj Biodiversity*, **3**, 3.
23. Thuile Bistarelli, L., Fuß, T., Walther, F., Zoccarato, L., **Talluto, L.**, Romaní, A. M., Grossart, H.-P. & Singer, G. A. (2024) Strong large-scale structure-function coupling in benthic bacteria is mediated by algae in a geodiverse river network. *Limnology and Oceanography*.
22. Weissman, J., Chappell, C., Oliviera, B., Evans, N., Fagre, A., Forsythe, D., Frese, S., Gregor, R., Ishaq, S., Johnston, J., Bittu, K., Matsuda, S., McCarren, S., Ortiz Alvarez de la Campa, M., Roepke, T., Sinnott-Armstrong, N., Stobie, C., **Talluto, L.**, Vargas-Muñiz, J. & The Advancing Queer and Trans Equity in Science (AQTES) Consortium (2024) Queer- and trans-inclusive faculty hiring – A call for change. *PLoS Biology*, **22**, e3002919.
21. Fandos, G., **Talluto, L.**, Fiedler, W., Robinson, R. A., Thorup, K. & Zurell, D. (2023) Standardised empirical dispersal kernels emphasise the pervasiveness of long-distance dispersal in European birds. *Journal of Animal Ecology*, **92**, 158–170.
20. Martini, J., Walther, F., Schenekar, T., Birnstiel, E., Wüthrich, R., Oester, R., Schindelegger, B., Schwingshackl, T., Wilfling, O., Altermatt, F., **Talluto, L.**, Singer, G. & Vitecek, S. (2023) The last hideout: Abundance patterns of the not-quite-yet extinct mayfly *Prosopistoma pennigerum* in the Albanian Vjosa River network. *Insect Conservation and Diversity*, **16**, 285–297.

19. Catalán, N., Campo, R. d., **Talluto, L.**, Mendoza-Lera, C., Grandi, G., Bernal, S., Schiller, D. v., Singer, G. & Bertuzzo, E. (2022) Pulse, Shunt and Storage: Hydrological Contraction Shapes Processing and Export of Particulate Organic Matter in River Networks. *Ecosystems*.
18. Knoflach, B., Ramskogler, K., **Talluto, L.**, Hofmeister, F., Haas, F., Heckmann, T., Pfeiffer, M., Piermattei, L., Ressler, C., Wimmer, M. H., Geitner, C., Erschbamer, B. & Stötter, J. (2021) Modelling of Vegetation Dynamics from Satellite Time Series to Determine Proglacial Primary Succession in the Course of Global Warming—A Case Study in the Upper Martell Valley (Eastern Italian Alps). *Remote Sensing*, **13**.
17. Thuile Bistarelli, L., Poyntner, C., Santín, C., Doerr, S. H., **Talluto, L.**, Singer, G. & Sigmund, G. (2021) Wildfire-Derived Pyrogenic Carbon Modulates Riverine Organic Matter and Biofilm Enzyme Activities in an In Situ Flume Experiment. *ACS ES&T Water*, **1**, 1648–1656.
16. Copenhaver-Parry, P. E., Carroll, C. J. W., Martin, P. H. & **Talluto, L.** (2020) Multi-scale integration of tree recruitment and range dynamics in a changing climate. *Global Ecology and Biogeography*, **29**, 102–116.
15. Münkemüller, T., Gallien, L., Pollock, L. J., Barros, C., Carboni, M., Chalmandrier, L., Mazel, F., Mokany, K., Roquet, C., Smyčka, J., **Talluto, L.** & Thuiller, W. (2020) Dos and don'ts when inferring assembly rules from diversity patterns. *Global Ecology and Biogeography*, **29**, 1212–1229.
14. Pollock, L. J., O'Connor, L. M. J., Mokany, K., Rosauer, D. F., **Talluto, L.** & Thuiller, W. (2020) Protecting Biodiversity (in All Its Complexity): New Models and Methods. *Trends in Ecology & Evolution*, **35**.
13. Vissault, S., **Talluto, L.**, Boulangeat, I. & Gravel, D. (2020) Slow demography and limited dispersal constrain the expansion of north-eastern temperate forests under climate change. *Journal of Biogeography*, **47**, 2645–2656.
12. Carboni, M., Guéguen, M., Barros, C., Georges, D., Boulangeat, I., Douzet, R., Dullinger, S., Klonner, G., Kleunen, M. van, Essl, F., Bossdorf, O., Haeuser, E., **Talluto, L.**, Moser, D., Block, S., Conti, L., Dullinger, I., Münkemüller, T. & Thuiller, W. (2018) Simulating plant invasion dynamics in mountain ecosystems under global change scenarios. *Global Change Biology*, **24**, e289–e302.
11. Haeuser, E., Dawson, W., Thuiller, W., Dullinger, S., Block, S., Bossdorf, O., Carboni, M., Conti, L., Dullinger, I., Essl, F., Klonner, G., Moser, D., Münkemüller, T., Parepa, M., **Talluto, L.**, Kreft, H., Pergl, J., Pyšek, P., Weigelt, P., Winter, M., Hermy, M., Van der Veken, S., Roquet, C. & Kleunen, M. van (2018) European ornamental garden flora as an invasion debt under climate change. *Journal of Applied Ecology*, **55**, 2386–2395.
10. **Talluto, L.**, Mokany, K., Pollock, L. J. & Thuiller, W. (2018) Multifaceted biodiversity modelling at macroecological scales using Gaussian processes. *Diversity and Distributions*, **24**, 1492–1502.
9. van Kleunen, M. et al. (2018) The changing role of ornamental horticulture in alien plant invasions. *Biological Reviews*, **93**, 1421–1437 [**L. Talluto** is author 28 of 29].
8. **Talluto, L.**, Boulangeat, I., Vissault, S., Thuiller, W. & Gravel, D. (2017) Extinction debt and colonization credit delay range shifts of eastern North American trees. *Nature Ecology and Evolution*, **1**, 0182.
7. Benkman, C. W., Jech, S. & **Talluto, L.** (2016) From the ground up: biotic and abiotic features that set the course from genes to ecosystems. *Ecology and evolution*, **6**, 7032–7038.
6. **Talluto, L.**, Boulangeat, I., Ameztegui, A., Aubin, I., Berteaux, D., Butler, A., Doyon, F., Drever, C. R., Fortin, M.-J., Franceschini, T., Liénard, J., McKenney, D., Solarik, K. A., Strigul, N., Thuiller, W. & Gravel, D. (2016) Cross-scale integration of knowledge for predicting species ranges: a metamodeling framework. *Global Ecology and Biogeography*, **25**, 238–249.

5. **Talluto, L.** & Benkman, C. W. (2014) Conflicting selection from fire and seed predation drives fine-scaled phenotypic variation in a widespread North American conifer. *Proceedings of the National Academy of Sciences of the United States of America*, **111**, 9543–9548.
4. **Talluto, L.** & Benkman, C. W. (2013) Landscape-scale eco-evolutionary dynamics: selection by seed predators and fire determine a major reproductive strategy. *Ecology*, **94**, 1307–1316.
3. Benkman, C. W., Fetz, T. & **Talluto, L.** (2012a) Variable resource availability when resource replenishment is constant: The coupling of predators and prey. *The Auk*, **129**, 115–123.
2. Benkman, C. W., Smith, J. W., Maier, M., Hansen, L. & **Talluto, L.** (2012b) Consistency And Variation In Phenotypic Selection Exerted By A Community Of Seed Predators. *Evolution*, **67**, 157–169.
1. **Talluto, L.** & Suding, K. N. (2008) Historical change in coastal sage scrub in southern California, USA in relation to fire frequency and air pollution. *Landscape Ecology*, **23**, 803–815.

Selected Work in Progress

3. Nakhavali, M., Lessa Derci Augustynczyk, A., Di Fulvio, F., Singh, R., Pruckner, S., **Talluto, L.** & Havlik, P. (2026) A global multi model dataset of climate extreme-event indicators for 1981-2100. *Scientific Data*, **in review**.
2. **Talluto, L.**, del Campo, R., Estévez, E., Fuss, T., Thuile-Bistarelli, L., Martini, J. & Singer, G. (2026) Modelling Biological and Resource Fluxes in Fluvial Meta-Ecosystems. *Methods in Ecology and Evolution*, **in review**.
1. Thuile-Bistarelli, L., Fuss, T., **Talluto, L.**, Masese, F. O. & Singer, G. A. (2026) Beyond discrete habitats: Benthic and planktonic lifestyles as emergent properties of one metacommunity. *The ISME Journal*, **in review**.

Presentations (Selected first-author)

13. Talluto, L., del Campo, R., Estévez, E., Fuß, T., Thuile Bistarelli, L. & Singer, G. A. (2022) "A model coupling dispersal and resource fluxes in fluvial meta-ecosystems". *SIL: Congress of the International Society of Limnology*.
12. Talluto, L. (2019) "Macroecology, data science, and the future of environmental management". *School of Biological Sciences, Queen's University Belfast*.
11. Talluto, L., Fuß, T. & Singer, G. A. (2019) "Modelling ecosystem metabolism at the scale of entire river networks". *Symposium for European Freshwater Sciences*.
10. Talluto, L. (2017a) Climatic disequilibrium in tree species distributions in North American forests. *Department of Evolutionary Biology and Environmental Studies, University of Zürich*.
9. Talluto, L. (2017b) Critical transitions and lagged response to climate change in eastern North American forests. *Alpine Ecology Lab, Grenoble, France*.
8. Talluto, L. (2017c) Improving biodiversity models by integrating multiple information sources. *Ecological Society of America Annual Meeting*.
7. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2017) Critical transitions and lagged response to climate change in eastern North American forests. *Department of Bioscience - Ecoinformatics and Biodiversity, Aarhus University, Denmark*.
6. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2016) Local dynamics slow the response of species ranges to climate change in eastern North American forests. *French Ecological Society Annual Meeting*.

5. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2015) Local colonization-extinction dynamics generate lags in the response to climate change in eastern North American forests. *American Geophysical Union Annual Fall Meeting*.
4. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2014a) A framework for cross-scale integration for predicting tree range shifts under climate change. *7th Eastern CANUSA conference in forest science*.
3. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2014b) A framework for cross-scale integration for predictive modeling of species' ranges. *Ecological Society of America Annual Meeting*.
2. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2014c) A state-transition approach to estimating the migration rate of the temperate-boreal forest transition under climate change. *Quebec Centre for Biodiversity Science Annual Symposium*.
1. Talluto, L., Boulangeat, I., Vissault, S. & Gravel, D. (2012) Effects of natural selection from seed predation on serotiny: The role of red squirrels in determining forest structure in lodgepole pine. *Ecological Society of America Annual Meeting*.

Mentoring

- 2025– **Adrienne Étard**, *Postdoc*, IIASA
- 2025– **Gemma Gerber**, *Postdoc*, IIASA
- 2025– **Elisa Stefaniak**, *Postdoc*, IIASA
- 2023–2024 **Antonia Dill**, *M.Sc.*, University of Innsbruck
- 2023–2024 **Lukas Prader**, *B.Sc.*, University of Innsbruck
- 2021–2022 **Franziska Walther**, *M.Sc.*, Humboldt-Universität Zu Berlin
- 2014–2016 **Steve Vissault**, *M.Sc.*, Université du Québec à Rimouski
- 2005 **Jocelyn Oakley**, *B.S.*, University of California, Irvine
- 2005 **Kimberly Kurcab**, *B.S.*, University of California, Irvine

Teaching Appointments

- 2020–2025 **Instructor**, University of Innsbruck, Innsbruck, Austria
Courses Taught: Bayesian Statistics (Ph.D.), Data Analysis (M.Sc), Macroecology (M.Sc), Field Ecology Excursion (M.Sc), Data Management and Statistics (B.Sc)
- 2018–2021 **Instructor**, IGB, Leibniz Institute for Freshwater Ecology and Inland Fisheries, Berlin, Germany
Courses Taught: Introductory Bayesian statistics for ecologists
- 2017–2018 **Guest Instructor**, *Department of Biology*, Sherbrooke University, Sherbrooke, Canada
Courses Taught: Bayesian Statistics for Ecologists (summer school)
- 2012 **Instructor**, *Department of Geography*, University of Wyoming, Laramie, Wyoming, USA
Courses Taught: Biogeography
- 2007 **Adjunct Professor**, *Department of Environmental Studies*, Mt. San Jacinto College, San Jacinto, California, USA
Courses Taught: Introduction to Environmental Studies
- 2005–2008 **Instructor**, *Department of Ecology and Evolutionary Biology*, University of California, Irvine, California, USA
Courses Taught: California Natural History; Experimental Biology Laboratory

2002–2005 **Teaching Assistant**, *Department of Ecology and Evolutionary Biology*, University of California, Irvine, California, USA
Courses Taught: Experimental Biology Laboratory; Processes in Ecology and Evolution; Patterns of Diversity, Ecology, and Evolution; Way Your Body Works (Human Physiology); COSMOS: High School Summer Program for Math and Science

Service & Professional Activities

2026– **Associate Editor**, *Methods in Ecology and Evolution*

2026– **Subject Editor**, *Ecography*

ongoing **Reviewing**

Nature Communications, Nature Ecology and Evolution, Agence Nationale de la Recherche (French National funding agency), Conservation Biology, Diversity and Distributions, Ecological Modelling, Ecology Letters, Ecography, Global Ecology and Biogeography, Journal of Biogeography, Journal of Ecology, Methods in Ecology and Evolution, NSF Division of Environmental Biology, PLoS One, Polish Academy of Sciences

2023–2024 **Workshop**, *Introduction to R*, University of Innsbruck, Biological Sciences Student Union

2023 **Neretva Science Week**, *Scientists for Balkan rivers*, <https://www.balkanrivers.net/en/neretva-science-week>

Participated as a researcher, focused on birds. Assisted with organisation of the event.

2017 **Symposium Co-organizer (with LJ Pollock and K Mokany)**, *New Approaches and Challenges for the Next Generation of Biodiversity Models*, Ecological Society of America Annual Meeting